

EXPLORER

OPEN EDITORS

main.c

MAIN.C

.vscode

main.dSYM

#include <stdio.h>

a.out

hello.c

main

main.c

sample.txt

tempCodeRunnerFile

Untitled-1

C main.c

main()

```
1 #include <stdio.h>
2
3 void printBinary(int n) {
4     for (int i = 7; i >= 0; i--) {
5         printf("%d", (n >> i) & 1);
6     }
7 }
8 int main() {
9     int num, shift;
10
11     printf("Enter a number (0-255): ");
12     scanf("%d", &num);
13
14     printf("Enter number of positions to shift: ");
15     scanf("%d", &shift);
16
17     printf("\nOriginal Number: %d\nBinary: ", num);
18     printBinary(num);
19
20     int leftResult = num << shift;
21     printf("\n\nLeft Shift (%d << %d) = %d\nBinary: ", num, shift, leftResult);
22     printBinary(leftResult);
23
24     int rightResult = num >> shift;
25     printf("\n\nRight Shift (%d >> %d) = %d\nBinary: ", num, shift, rightResult);
26     printBinary(rightResult);
27
28     printf("\n");
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

zsh

+

-

+

-

...

+

-

abhaygupta@192 main.c % gcc main.c

abhaygupta@192 main.c % ./a.out

Enter a number (0-255): 30

Enter number of positions to shift: 4

Original Number: 30

Binary: 00011110

Left Shift (30 << 4) = 480

Binary: 11100000

OUTLINE

TIMELINE

Experiment 11.2

```
#include <stdio.h>
```

```
void PrintBinary (int n) {  
    for (int i=7; i >= 0; i--) {  
        printf ("%d", (n >> i) & 1);  
    }  
}
```

```
int main () {  
    int num, shift;  
    printf ("Enter a number (0-255):");  
    scanf ("%d", &num);
```

```
    printf ("Enter number of positions to shift:");  
    scanf ("%d", &shift);
```

```
    printf ("In Original Number : %d \n Binary :", num);  
    PrintBinary (num);
```

```
    int leftResult = num << shift;  
    printf ("\n\n Left shift (%d << %d) = %d \n Binary:", num, shift,  
        leftResult);  
    PrintBinary (leftResult);
```

Teacher's Signature : _____

```
int rightResult = num >> shift ;  
printf ("In Right shift (%d >> %d) = %d\nBinary:", num,  
        shift, rightleftResult);  
PrintBinary (leftrightResult);  
  
Print ("In")  
return 0;  
}
```